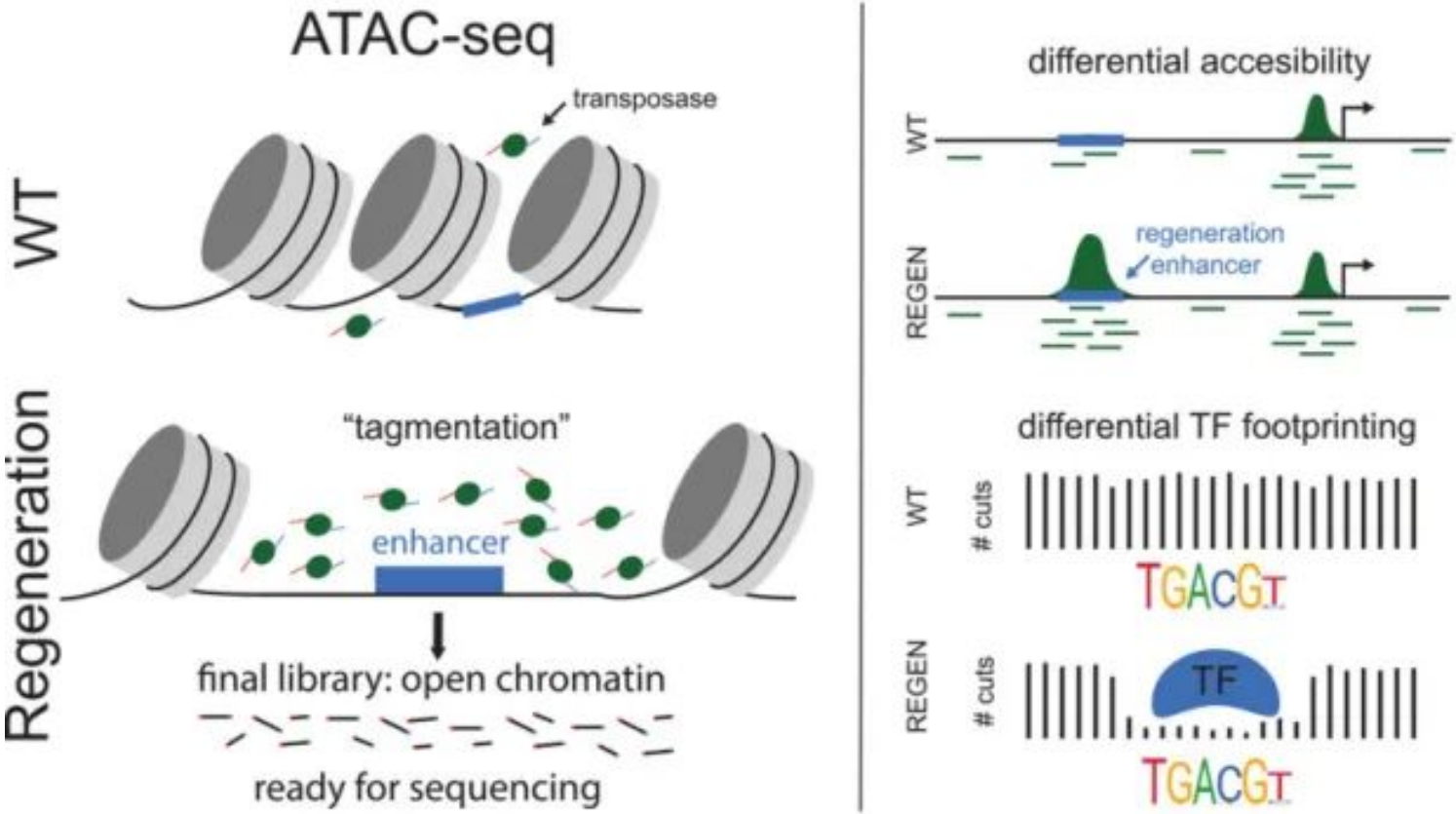
ATAC-seq (Assay for Transposase-Accessible Chromatin using sequencing)

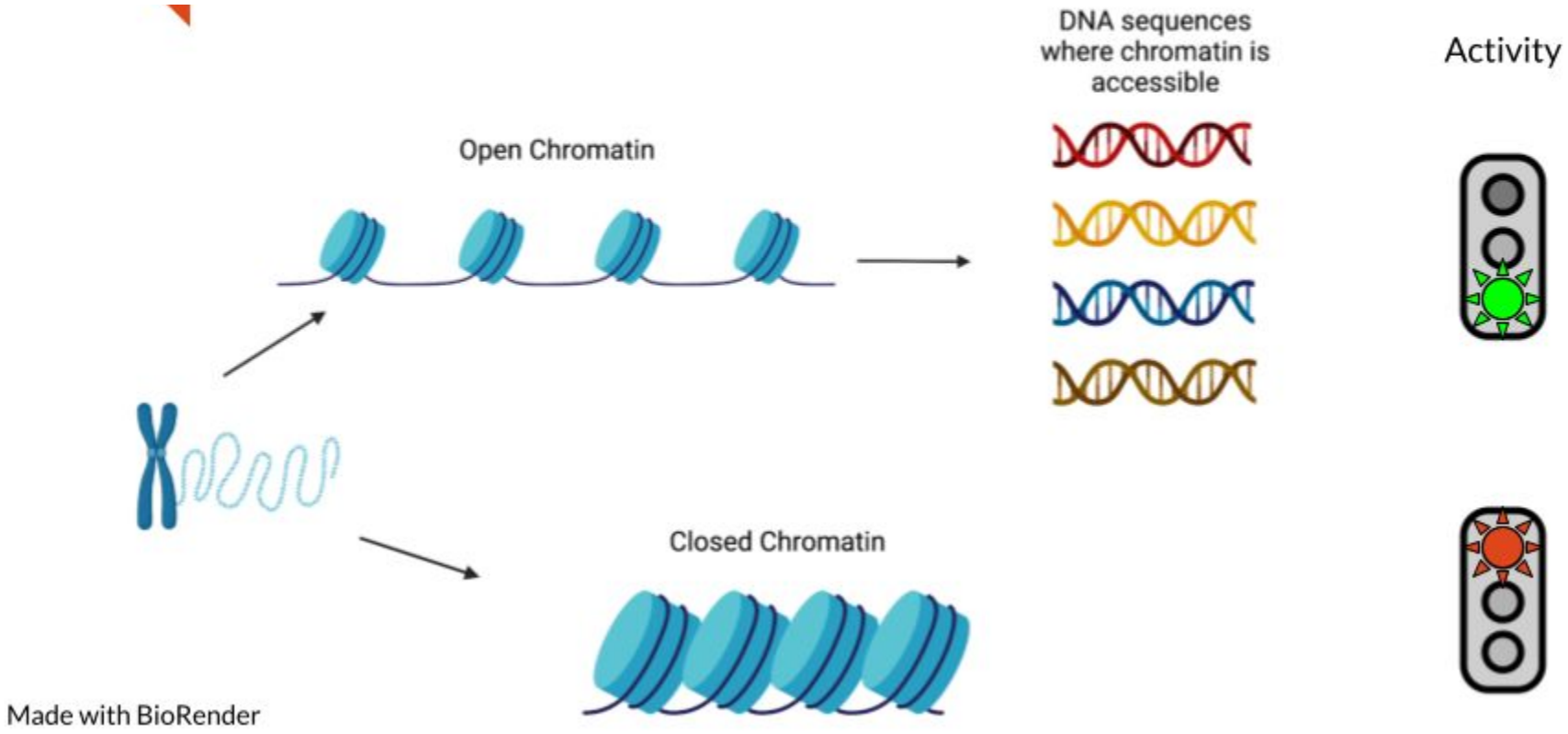
Quynh Nhu Nguyen

What is ATAC-seq?



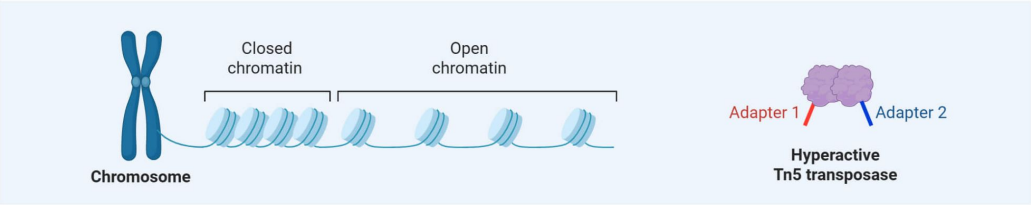
https://www.ncbi.nlm.nih.gov/books/NBK586944/figure/ch29.Fig1/

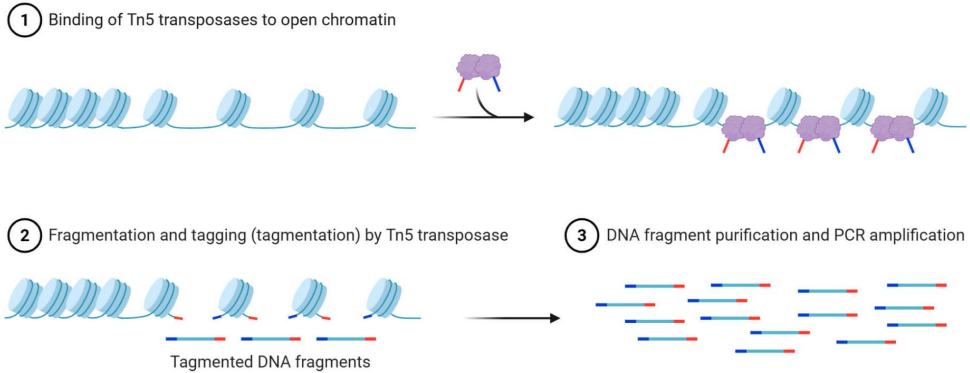
What does accessibility to chromatin represent?

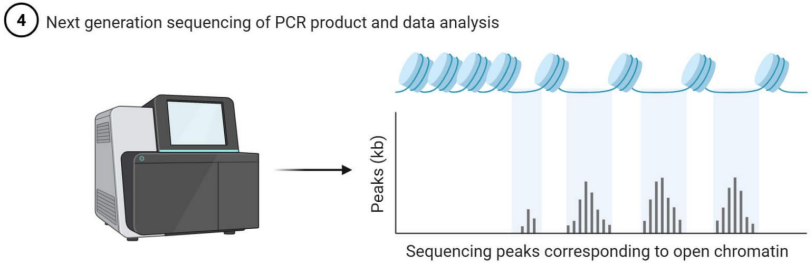


https://hutchdatascience.org/Choosing_Genomics_Tools/atac-seq-1.html

Steps of ATAC Sequencing

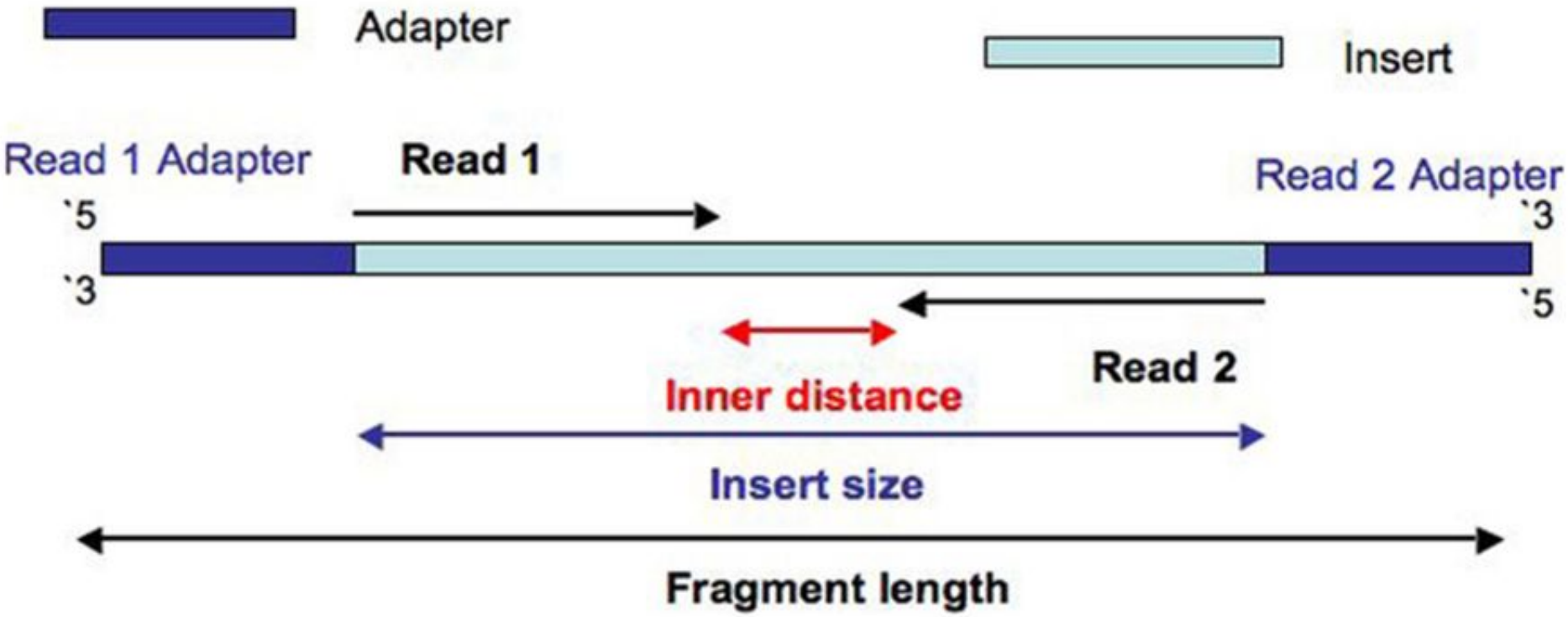






https://microbenotes.com/atac-seq/

Library Fragment Classification by Insert Size



https://www.biostars.org/p/106291/

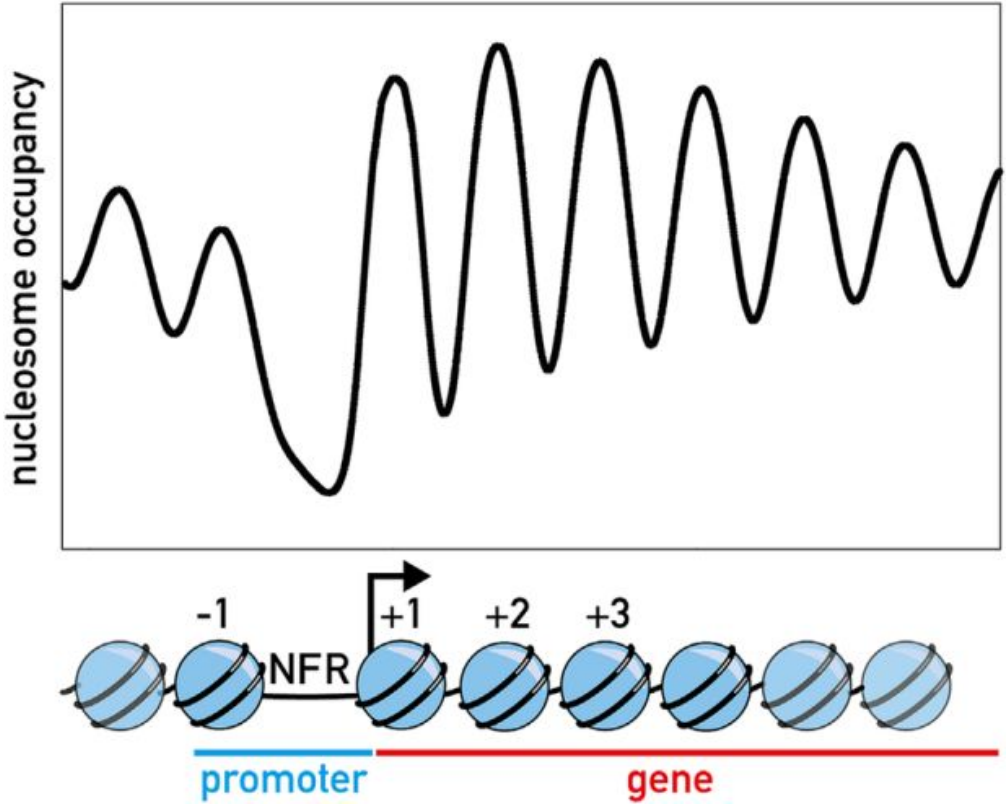
Nucleosome-free fragments

50–100 bp: Nucleosome-free regions

150–200 bp: Mono-nucleosome regions

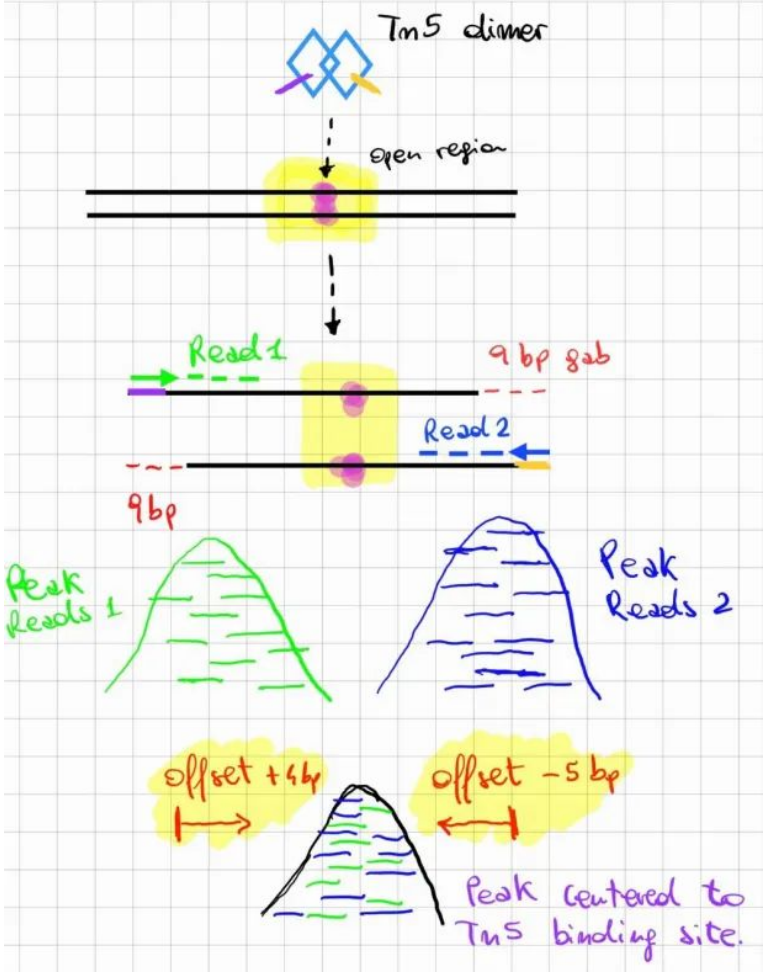
300–400 bp: Di-nucleosome regions

600 bp: Tri-nucleosome regions



https://bioinformaticamente.com/2025/01/01/comprehensive-guide-to-atac-seq-data-analysis/

Read Shifting



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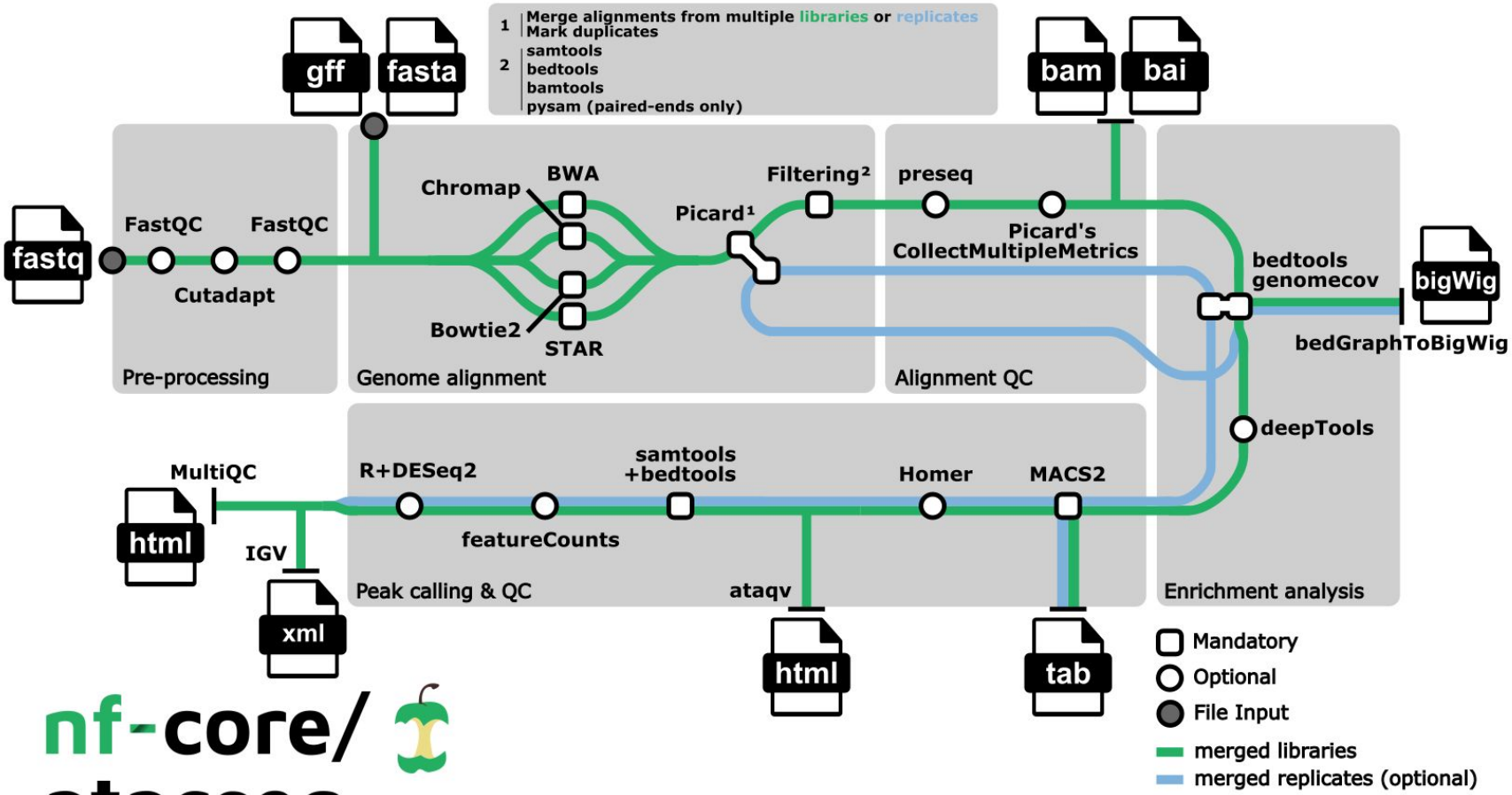
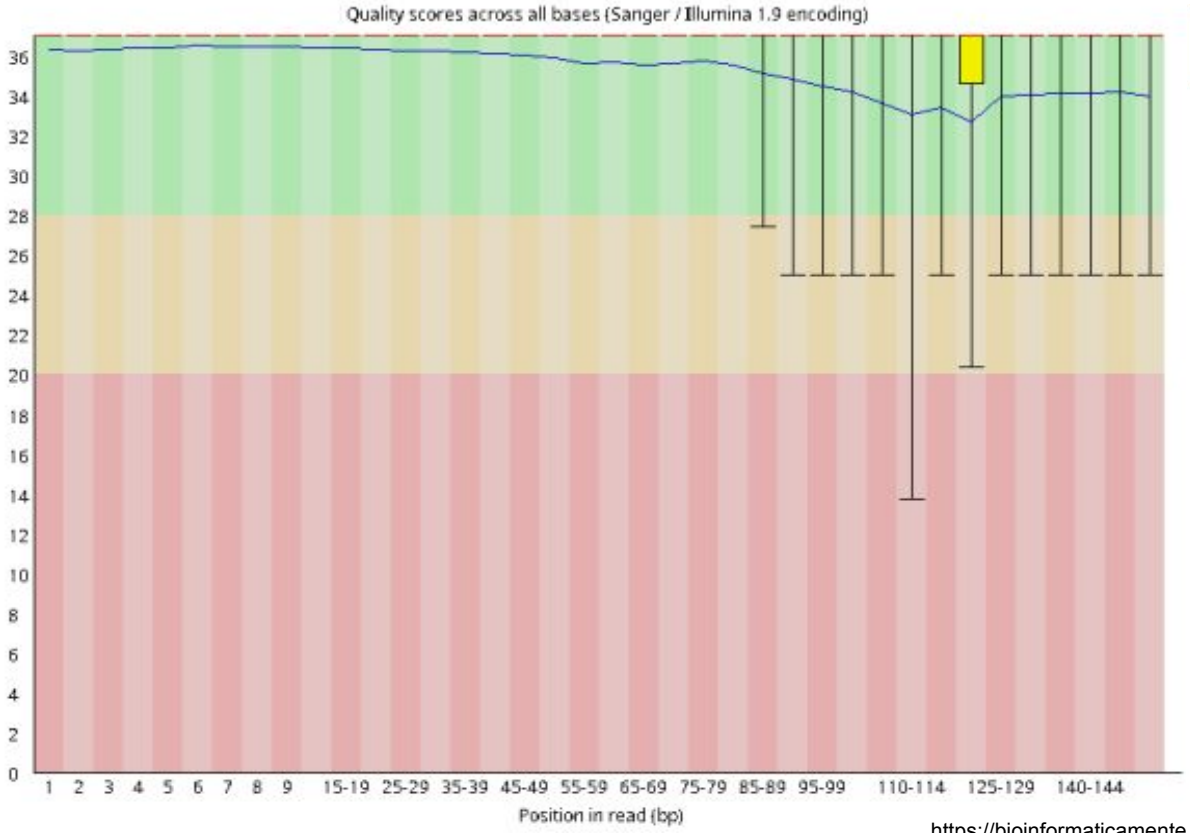
ATACseq pipeline



Read quality

Per base sequence quality




```
+SEQ_ID
!''*((((***+))%%%++)(%%%%).1**
```

A quality value Q is an integer representation of the probability p that the corresponding base call is incorrect.

$$Q = -10 \log_{10} P \quad \Longrightarrow \quad P = 10^{\frac{-Q}{10}}$$

Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10000	99.99%
50	1 in 100000	99.999%

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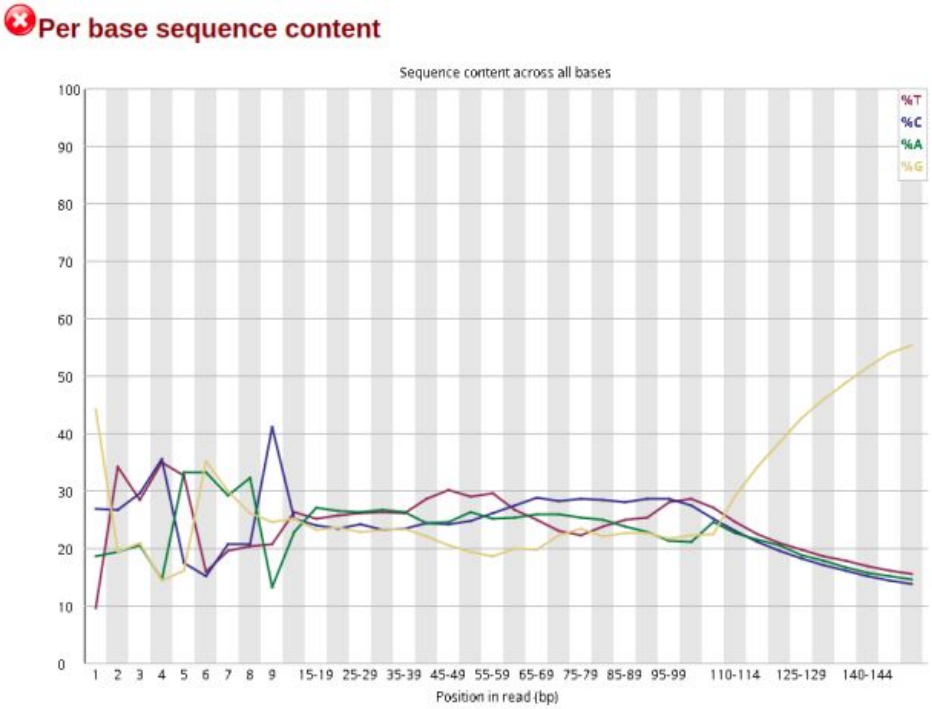
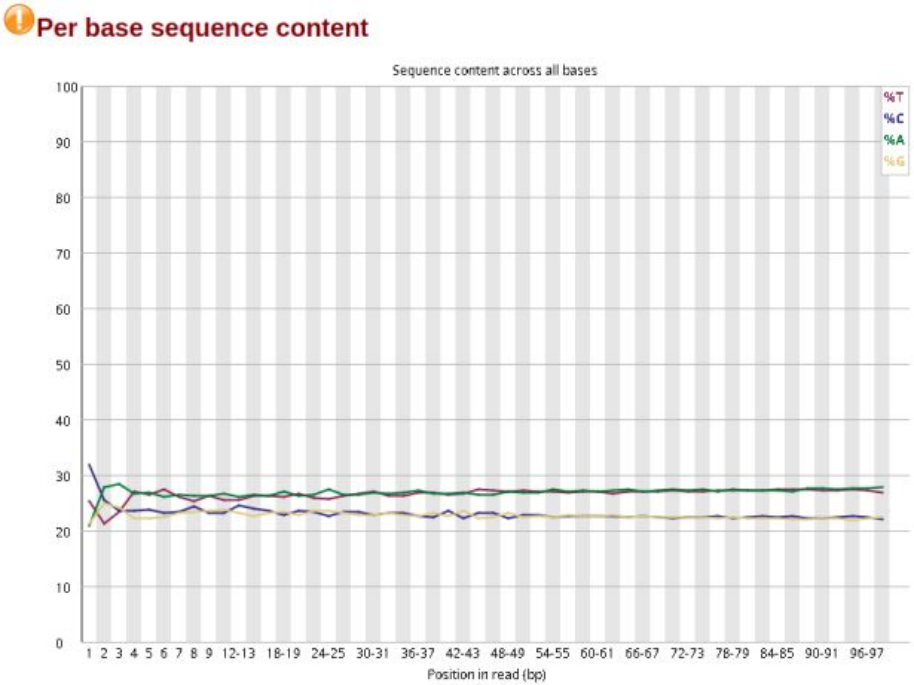
Per base sequence content





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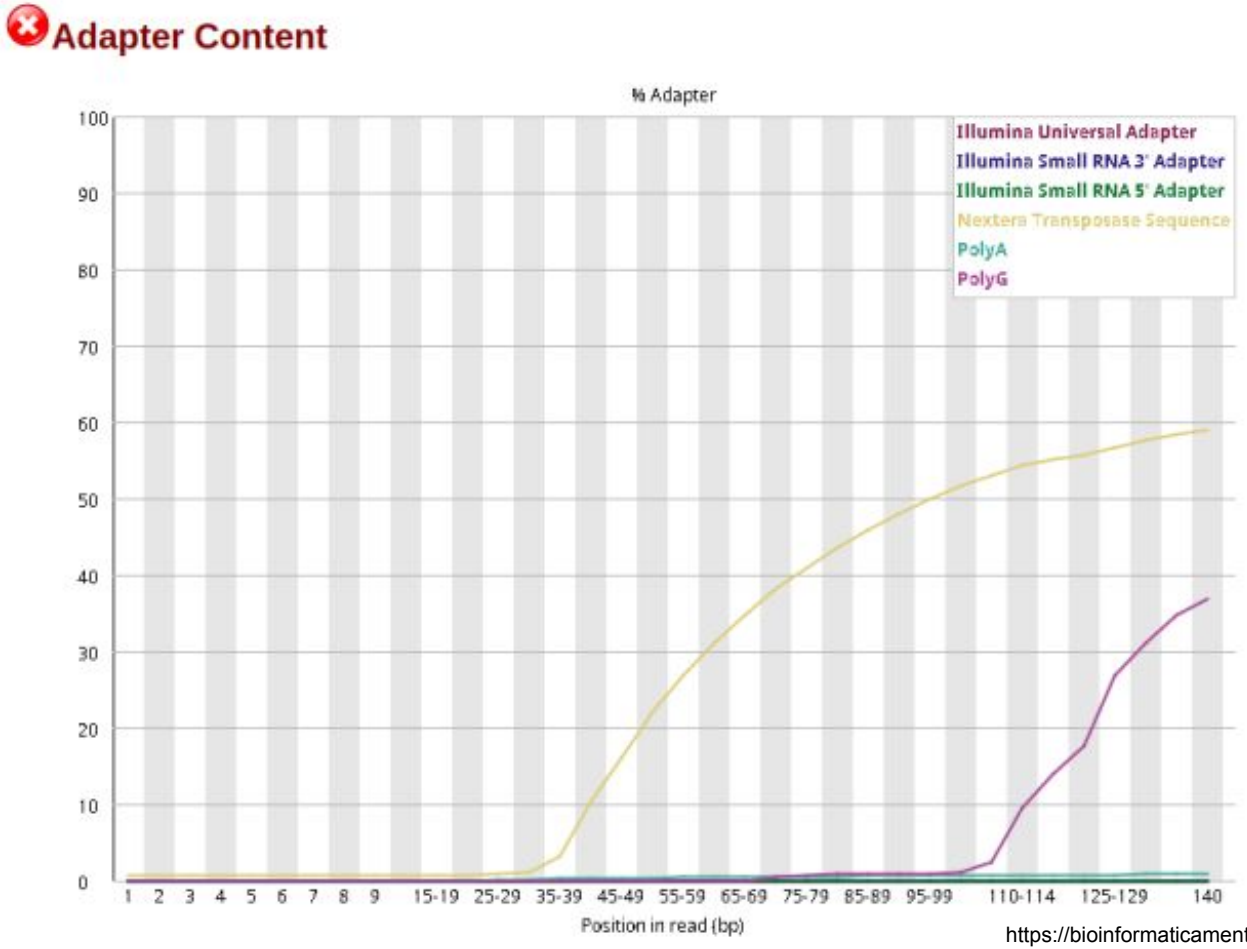
Adapter Content



https://bioinformaticamente.com/2025/01/01/comprehensive-guide-to-atac-seq-data-analysis/

Overrepresented Sequences

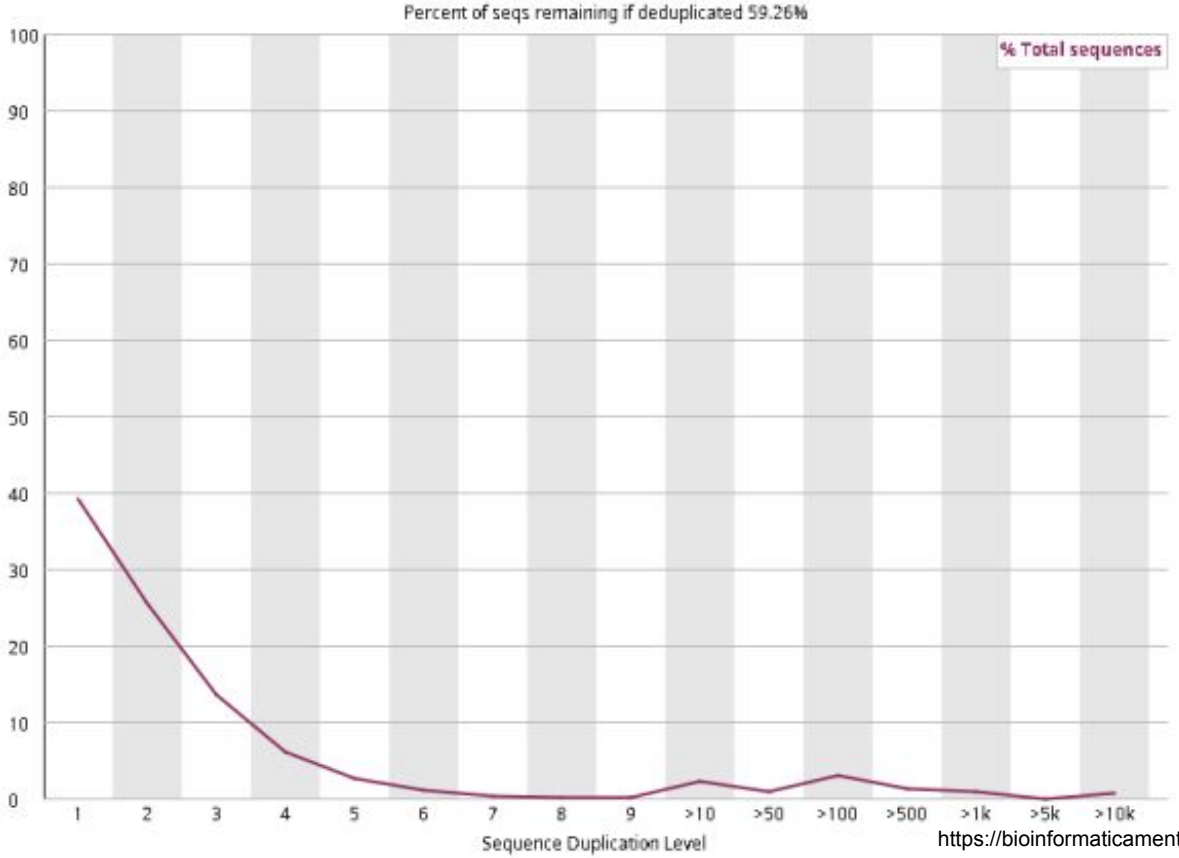
Overrepresented sequences

Sequence	Count	Percentage	Possible Source
CTGTCTCTTATACACATCTCCGAGCCCACGAGACCGGAGCCTATCTCGTA	272183	0.5291201129658407	No Hit
CTGTCTCTTATACACATCTCCGAGCCCACGAGACCGGAGCCTATCGCGTA	127369	0.24760363310106126	No Hit

https://bioinformaticamente.com/2025/01/01/comprehensive-guide-to-atac-seq-data-analysis/

Duplicate Reads

Sequence Duplication Levels



https://bioinformaticamente.com/2025/01/01/comprehensive-guide-to-atac-seq-data-analysis/

Alignment / Mapping Quality

Mapping rate (%)

Fraction of reads successfully aligned to the reference genome

Typical good: >70–80% (depends on sample quality)

Uniquely mapped reads (%)

Reads mapped to a single genomic location

Important for ATAC-seq signal specificity

Multi-mapped reads (%)

High values may indicate repetitive regions or poor library complexity

Mapping Quality Score (MAPQ)

Often filter with: MAPQ ≥ 30

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Read Duplication & Library Complexity

Duplicate rate (%)

- From PCR duplicates (e.g., using Picard MarkDuplicates)
- ATAC-seq acceptable range: varies, but lower is better (<30–50%)

Library complexity

- Estimated via tools like preseq
- Indicates diversity of fragments

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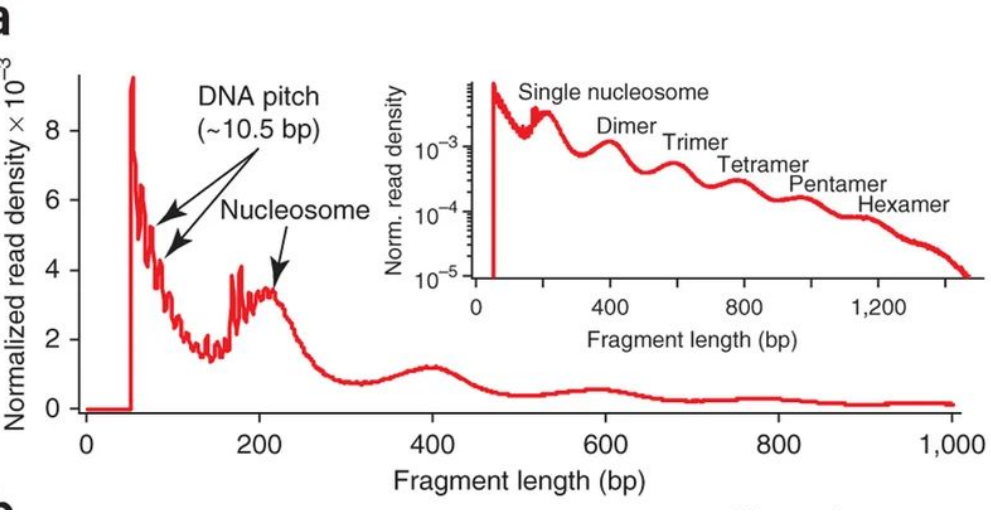
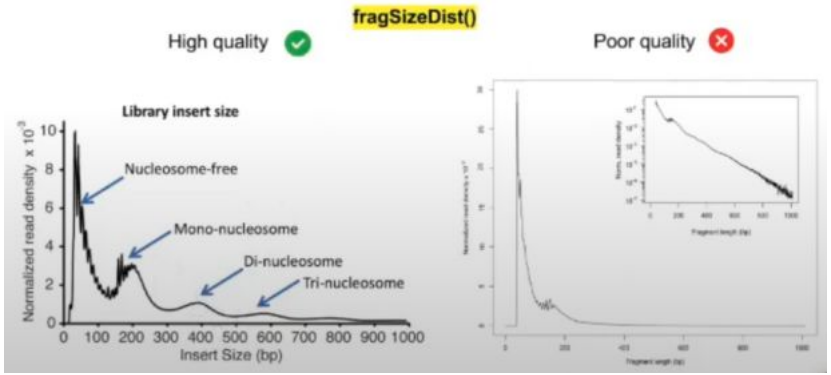
Insert Size Histogram





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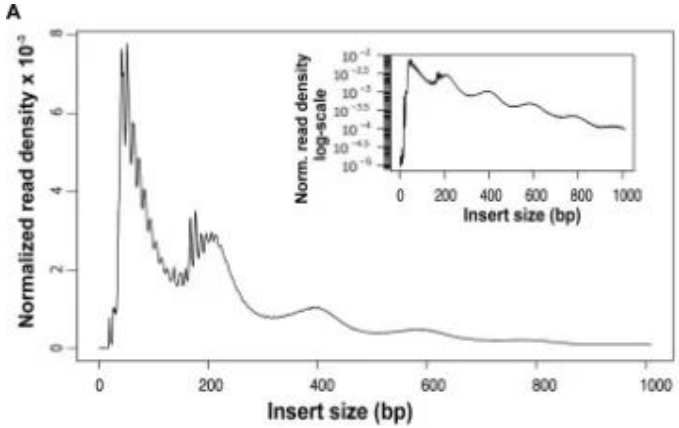
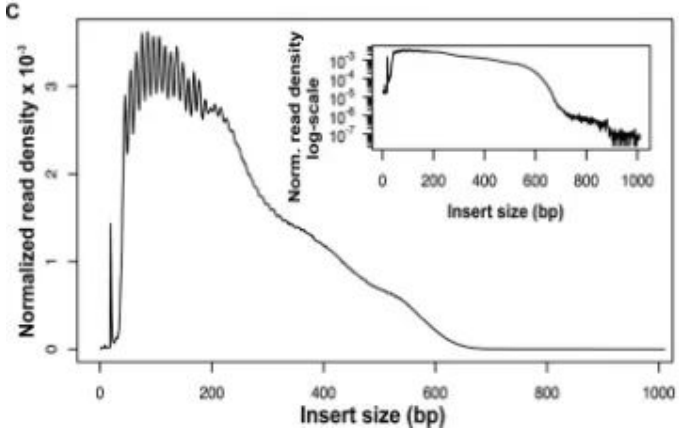
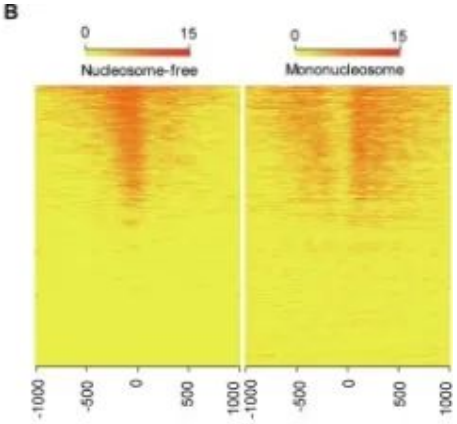
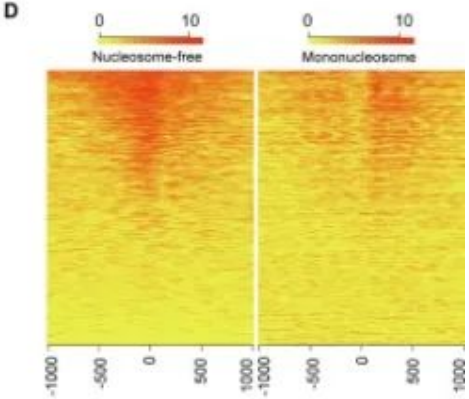
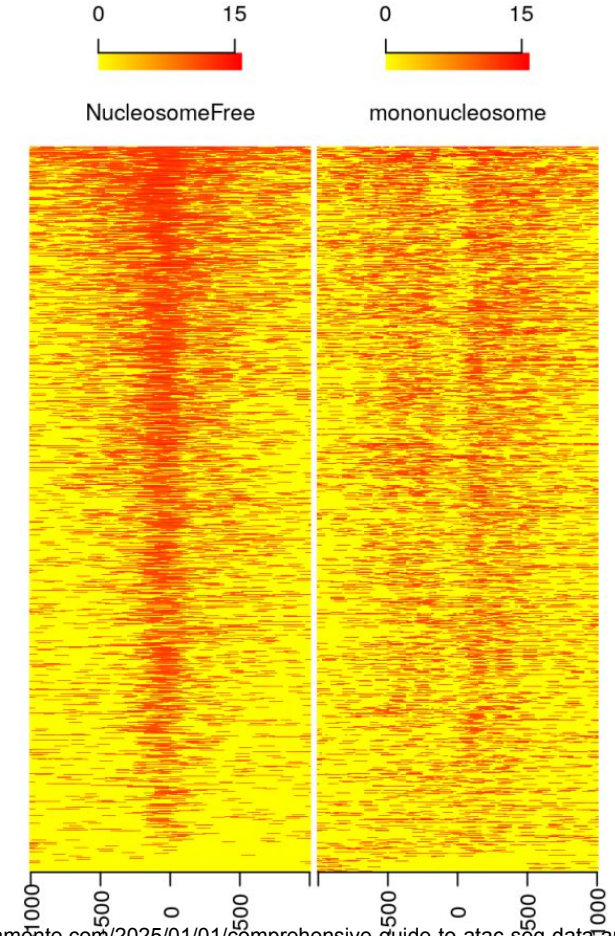
Insert Size Histogram











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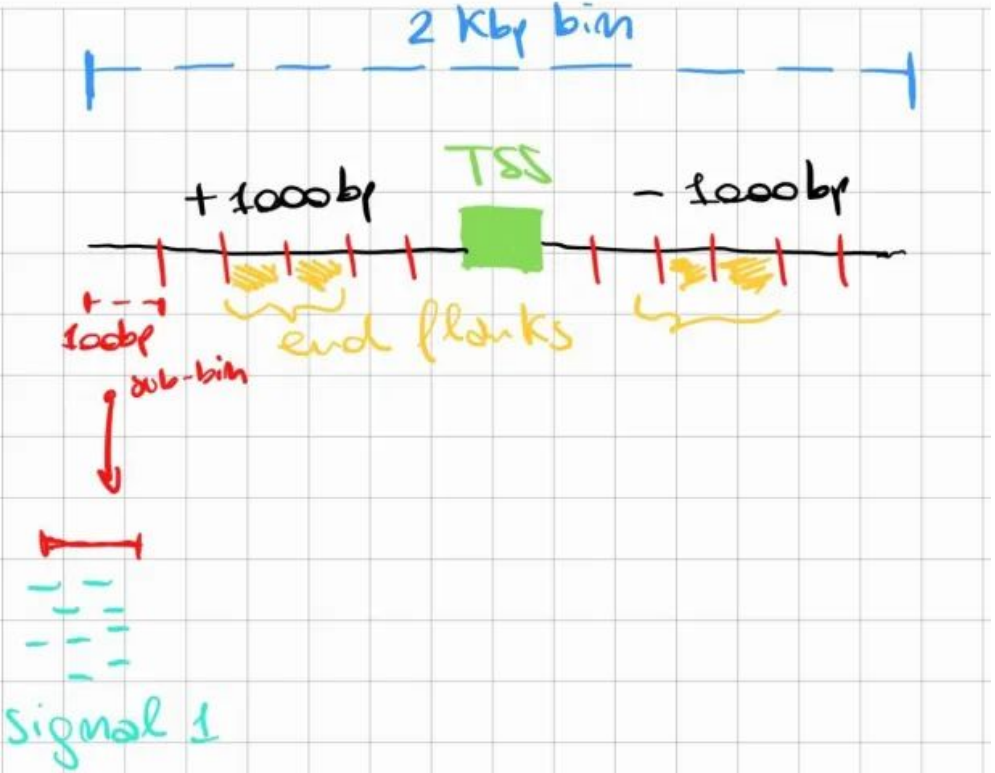
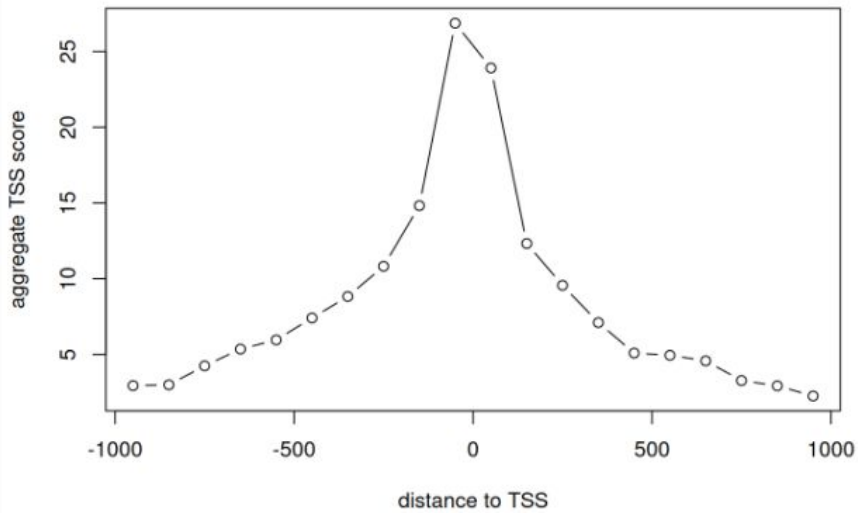
TSS Enrichment Score



$$\text{TSS score} = \frac{\text{max signal in 2Kbp bin}}{\text{mean signal in 2Kbp bin}}$$



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Fraction of Reads in Peaks (FRiP)

Fraction of mapped reads that fall into called peaks

Interpretation:

>0.2 (20%) → good, Lower values → noisy data

Suppose you have:

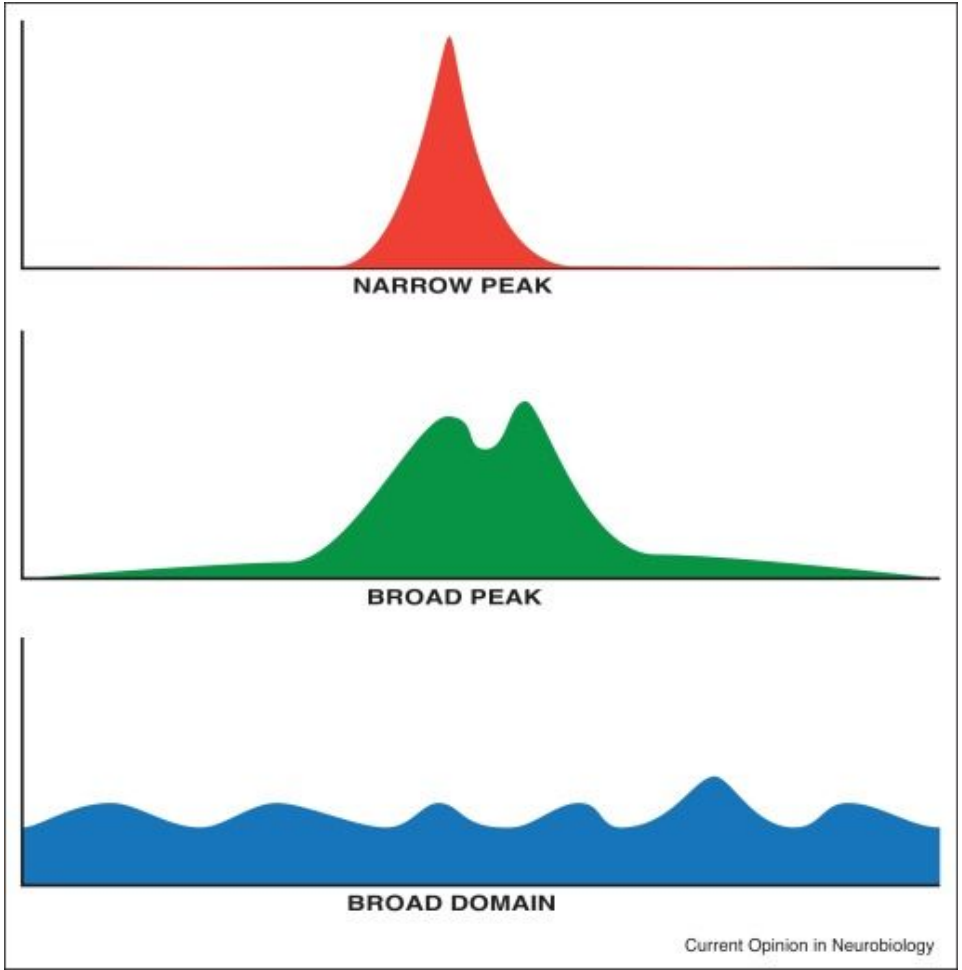
- Total mapped reads = **40,000,000**
- Reads overlapping peaks = **10,000,000**

Then:

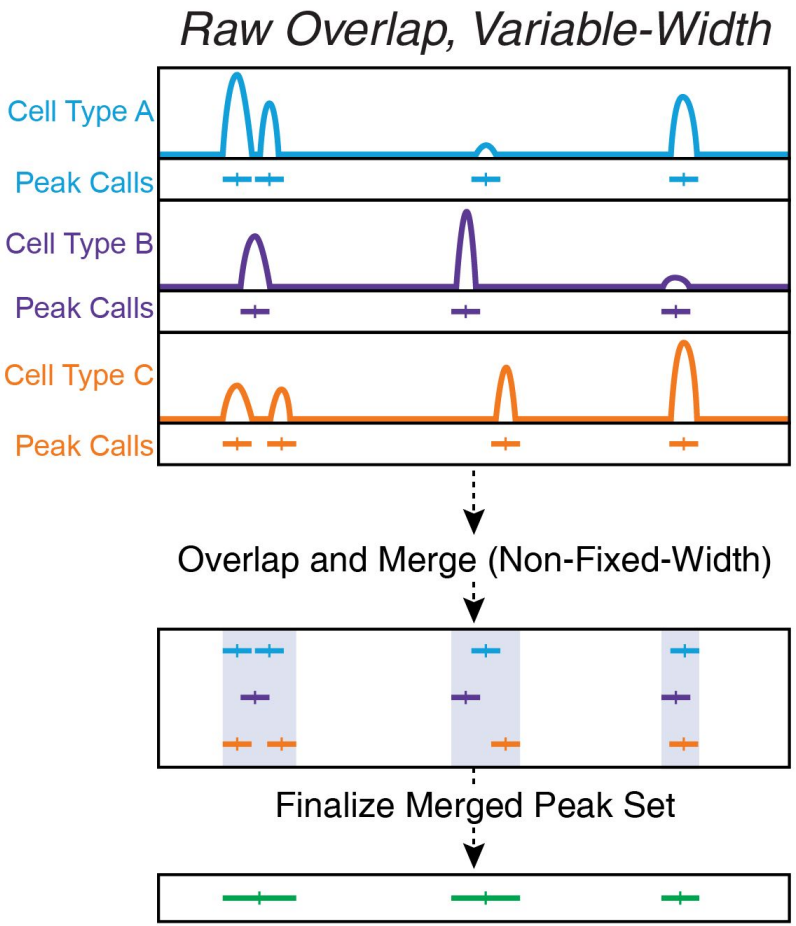
FRiP=10,000,000 / 40,000,000 = 0.25

👉 **FRiP = 0.25 (25%) → good quality**

Peaks


NARROW PEAK
BROAD PEAK
BROAD DOMAIN
Current Opinion in Neurobiology

Consensus peaks



https://www.archrproject.com/archive/1.0.1/bookdown/the-iterative-overlap-peak-merging-procedure.html